

Zhimeng Wang

PhD Student in Electrical and Computer Engineering, Johns Hopkins University

[Website][LinkedIn][Google Scholar]

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Education

- **Johns Hopkins University**

Aug. 2024 - 2029 (prospective)

- PhD Student in Electrical and Computer Engineering
- Advisor: Prof. Enrique Mallada.
- Research topic: Decentralized stability certificates for power grids with high renewable penetration

- **Tsinghua University**

Sept. 2021 - Jun. 2024

- M.S. in Electrical Engineering GPA: 3.84/4.00
- Advisors: Prof. Hongbin Sun, Prof. Xinwei Shen.
- Thesis: Robust planning for offshore microgrid considering tidal current power, desalination and frequency stability

- **University of California, Berkeley**

Aug. 2019 - Dec. 2019

- Exchange student
- Course works: Introduction to Electric Power Systems, Optimization Models in Engineering, etc.

- **Beijing University of Chemical Technology**

Sept. 2017 - Jun. 2021

- B.Eng. in Automation, ranked 7/153 (top 5%) GPA: 88.73/100
- Graduation project: Research on image motion deblurring methods based on deep learning

Publications

[3] **Zhimeng Wang**, Sushobhan Chatterjee, Sijia Geng, Richard Pates, Enrique Mallada, “Decentralized stability certificates in IBR-dominated grids: the role of the network state”, 2026, *PowerUp*. [PDF]

- Propose a model to analyze the stability of IBR-dominated grids.
- Propose and analyze decentralized stability certificates in terms of network states.

[2] **Zhimeng Wang**, Ang Xuan, Xinwei Shen, Yunfei Du, Hongbin Sun, “A robust planning model for offshore microgrid considering tidal power and desalination”, *Applied Energy*, 2023, 350: 121713. [PDF]

- A two-stage robust model of an offshore microgrid considering tidal power generation and seawater desalination is proposed.
- Uncertainty set of tidal power generation is defined in terms of the tidal level.
- Scenarios with different tidal delay is simulated.
- Complementarity between tidal power generation and the other renewable energy sources is proved.

[1] Jiajia Huan, **Zhimeng Wang**, Yunfei Du, Xinwei Shen, Baihao Qiao, Chungeng He, Xiaodong Lan, Shuxin Luo, “Boundary Inference of Load Scenarios in Multi-energy Parks Based on Statistical Learning”, *2023 5th International Conference on Power and Energy Technology (ICPET)*, Tianjin, China, 2023, pp. 1530-1535. [PDF]

- The paper was selected as the “**best student paper finalist**” and I delivered an **oral speech**.

Awards and Scholarships

- **Carl E. Heath Fellowship**, Johns Hopkins University *Aug. 2024 - Jul. 2025*
- **First Class Scholarship**, Tsinghua University *Oct. 2023*
 - Awarded to around top 2% students with best overall performance in the previous academic year.
- **Second Class Scholarship**, Tsinghua University *Oct. 2022*
 - Awarded to around top 20% students with best overall performance in the previous academic year.
- **Beijing Outstanding Graduates**, Beijing Ministry of Education *Jun. 2021*
 - Awarded to top 5% of graduates.
- **Second Prize, National Physics Competition**, Beijing Physics Society *Dec. 2018*
 - Ranked #1 among all the participants from my university.

Teaching

- **Teaching Assistant of EN520.637 Foundations of Reinforcement Learning**, Johns Hopkins University *Fall 2025, Fall 2026*
 - Host recitation sessions and office hours, grade assignments.

Industrial Projects

- **Research on Energy Efficiency Data Mining Technology for Multi-energy Complementary Park Planning** *Sept. 2021- Jun. 2023*
 - A cooperative project between China Southern Power Grid Guangdong Power Grid Co. LTD and Shenzhen International Graduate School, Tsinghua University.
 - The project focuses on giving benefit analysis of park-level integrated energy system transformation planning based on electricity substitution.
 - **My contributions:** **Conducted** the part of the project of providing boundary of uncertainty sets of the load scenarios with no request for former data; **Submitted a patent** titled “Method, Device, Terminal Equipment, and Storage Medium for Determining the Load Boundary of Multi-energy Parks”; **Published a conference paper** titled “Boundary Inference of Load Scenarios in Multi-energy Parks Based on Statistical Learning”; **Drafted research reports**; Wrote part of the **back-end code** for the project software.
- **Research on Stochastic Programming Methods of Integrated Energy System Considering Operation** *Jun. 2021 - Sept. 2021*
 - This is a cooperative project between Electric Power Research Institute, China Southern Power Grid and Shenzhen International Graduate School, Tsinghua University.
 - The project focuses on providing integrated energy systems with a planning method while considering uncertainties in a stochastic programming manner.
 - **My contributions:** **Wrote and debugged** the codes; **Drafted research reports**.